An Introduction to Cryptography

Course Syllabus

Cypher Stack Academy · Online, Live Lectures · 13 Weeks

November 2026 – January 2027

Last updated: October 17, 2026

Notice to Students

This course is offered independently by **Cypher Stack**, not through an accredited academic institution. It does not carry academic credit, and no certificate of accreditation is issued on completion. **Lectures are free and attendance is optional. Office hours, graded problem sets, and one-on-one sessions are offered through paid enrollment tiers**.

About Cypher Stack

Cypher Stack (cypherstack.com) is a multidisciplinary firm that offers design, development, research, and infrastructure services to clients around the world. The research arm is a team of mathematicians and cryptographers who have completed projects for several prominent groups such as the Monero Research Lab and the Litecoin Foundation.

This course grows out of that work. The mathematics is taught from first principles, but the people teaching it use these tools professionally: the discrete logarithm problem, elliptic curves, and probabilistic analysis covered in this syllabus are the same tools behind the team's own research.

Selected Recent Work

The instructors are among the authors of the following, all publicly available.

- B. Goodell, R. Salazar, F. Slaughter & L. Szramowski, "SLVer Bullet: Straight-Line Verification for Bulletproofs," Cryptology ePrint Archive, Paper 2025/1345. eprint.iacr.org/2025/1345
  Develops a framework for proof of computation over elliptic curves, with full protocol descriptions and proofs of completeness and soundness, and applies it to Schnorr and Bulletproofs to substantially reduce verification cost. *Related to Weeks 4–6 and 13.*

- B. Goodell, R. Salazar & F. Slaughter, "Uniformly Most Powerful Tests for Ad Hoc Transactions in Monero," Cryptology ePrint Archive, Paper 2025/221. eprint.iacr.org/2025/221
  Constructs an optimal statistical test, via the Neyman–Pearson Lemma, that classifies Monero transactions by how they were built, and uses it to describe a maximum-likelihood de-anonymization attack. *Related to Week 10.*

- B. Goodell, R. Salazar, F. Slaughter & L. Szramowski, *The MWEB Grimoire: A Comprehensive Mathematical Overview of Litecoin's Mimblewimble Extension Blocks*, Cypher Stack, 2026. github.com/cypherstack/mweb-grimoire
  Fills a gap in the documentation of MWEB as deployed in Litecoin. Builds the protocol from its primitives (Pedersen commitments, range proofs, Schnorr signatures, and Merkle mountain ranges), specifies stealth addresses, kernels, and the peg-in and peg-out processes, and analyzes its security guarantees, including inflation resistance, theft resistance, and the horizon attack. *Related to Weeks 4, 11, and 13.*

Course Instructors

Instructor	**Rigo Salazar**, M.S. in Mathematics
E-mail	**rigo@cypherstack.com**
Instructor	**Luke Szramowski**, M.S. in Mathematics (x2)
E-mail	**luke@cypherstack.com**
Office hours	Four virtual sessions per week for students enrolled in Tier 2 or Tier 3 (see *Enrollment Tiers* below)
One-on-one	Two 30-minute sessions per month for students enrolled in Tier 3
Platform	Lectures: on a platform that will be announced once decided. Office hours: Jitsi or Signal.

Course Description

The aim of this course is to familiarize students with cryptographic concepts, beginning with the fundamental mathematical principles and proceeding onward to their application in modern implementations. The course is meant to be accessible to people of all backgrounds. Topics run from divisibility, modular arithmetic, and proof technique, through groups, rings, and finite fields, to Diffie–Hellman, the discrete logarithm problem, lattices and Learning with Errors, primality testing, security definitions, RSA, and elliptic curve cryptography.

Prerequisites

The course is structured to be a 'first-principles' style class. Every topic that will be needed is contained within the course. However, since the concepts, even when simplified, are still difficult, a student would benefit from strong reasoning and algebra skills.

Format and Methods of Instruction

- **Live online lectures.** The course is neither face to face nor asynchronous. The lectures will be held on **[a platform that will be announced soon]**. Each week, the same lecture will be given at two different times to accommodate students' schedules, so that students in different time zones can attend live. Students need only attend one of the two meetings.
- **Time Commitment.** The course will take place over 13 weeks, with 1 lecture a week, lasting 1 hour. Students will be given optional additional reading and problem sets to supplement the lectures.
- **Problem sets.** One problem set accompanies each week's lecture to supplement learning.
- **Office hours.** Held virtually four times per week for students enrolled in Tier 2 or Tier 3 (see below).
- **One-on-one sessions.** Available to students enrolled in Tier 3 (see below).

Meeting Times

Session A	**[day, time, and time zone]**
Session B	**[day, time, and time zone]**

Enrollment Tiers, Attendance, and Office Hours

- **Lectures are entirely free.** There is no fee to register for or attend the lectures.
- **Attendance is optional.** No attendance is taken toward any grade.
- **Additional support is offered in tiers.** Students who want direct interaction with the instructors may enroll in one of two paid tiers, summarized below. Each tier includes every benefit of the tier before it.

Tier	Includes	Cost
Tier 1	Live lectures, weekly problem sets, and supplementary reading	Free
Tier 2	All Tier 1 benefits, plus access to weekly office hours and hand grading of submitted problem sets	$600
Tier 3	All Tier 2 benefits, plus two 30-minute one-on-one sessions per month with an instructor	$1,500

*Fees cover the full 13-week course. Payment and registration details are **[to be announced]**.*

Tier 1: Lectures

Tier 1 is open to everyone at no cost. Students in this tier may attend either weekly lecture session and receive all problem sets and supplementary material. Problem sets submitted by Tier 1 students are not graded.

Tier 2: Office Hours and Graded Problem Sets

Office hours are held virtually via Jitsi or Signal **four times per week**, with each session lasting approximately one hour. At the start of the course, students enrolled in Tier 2 or Tier 3 will be asked to submit their weekly availability. The instructors will then schedule the office hours to accommodate as many students as possible, spreading the sessions as evenly as possible across the week. Students in this tier may also submit problem sets to be graded by hand, with written feedback from the instructors.

Tier 3: One-on-One Sessions

In addition to all Tier 2 benefits, students enrolled in Tier 3 may schedule **two 30-minute one-on-one sessions per month** with an instructor. These sessions are held virtually and may be used for any course-related purpose, such as reviewing difficult material, working through problem sets, or exploring topics beyond the lectures. Sessions are arranged directly with the instructors at

a mutually convenient time.

Methods of Evaluation

There are no examinations and no course grade.

- **Problem Sets.** Problem sets are optional. They are provided to every student, but they are graded only for students enrolled in Tier 2 or Tier 3.
- **Students enrolled in Tier 2 or Tier 3** may submit problem sets to be graded by hand, with written feedback. Submission format and turnaround are **[to be announced]**.

Course Materials

No textbook purchase is required; every topic that will be needed is contained within the course. Each lecture and each problem set ends with its own Sources box, and a consolidated reference list collecting every source named across the course will be provided separately. For readers who want one book for the whole course, the following are suggested (free resources marked).

For the whole course — pick one

- V. Shoup, *A Computational Introduction to Number Theory and Algebra*, 2nd ed. — closest single match to the syllabus. **Free:** https://shoup.net/ntb/
- J. Hoffstein, J. Pipher & J. H. Silverman, *An Introduction to Mathematical Cryptography*, 2nd ed., Springer — gentler; aimed at the second half.
- J. Katz & Y. Lindell, *Introduction to Modern Cryptography*, 3rd ed. — the standard reference for security definitions.
- D. Boneh & V. Shoup, *A Graduate Course in Applied Cryptography* — harder; the next step after this course. **Free:** https://toc.cryptobook.us/

Free supplementary resources

- R. Hammack, *Book of Proof*, 3rd ed. — proof methods and quantifiers. https://www.people.vcu.edu/~rhammack/BookOfProof/
- T. Judson, *Abstract Algebra: Theory and Applications* — groups, rings, and finite fields. http://abstract.pugetsound.edu/
- A. Menezes, P. van Oorschot & S. Vanstone, *Handbook of Applied Cryptography* — dated in places; best single algorithms reference. https://cacr.uwaterloo.ca/hac/

Student Learning Outcomes

The course moves from fundamental mathematical principles to their application in modern implementations. By the end of the course, students who engage with the lectures and supplementary material will be able to:



- Read and write the mathematical notation and proof methods used in the analysis of cryptography.
- Carry out arithmetic in $\mathbb{Z}_n$ and in finite fields, including the Euclidean Algorithm, modular inverses, and the Chinese Remainder Theorem.
- Describe groups, rings, and fields, and use Lagrange's Theorem and primitive roots to reason about cryptographic constructions.
- Explain why the Diffie–Hellman key exchange and RSA work, and identify the hard problems they rely on.
- Describe standard attacks, including Baby-Step Giant-Step, Pollard's Rho, and attacks on RSA, and what they cost.
- Explain the Learning with Errors problem and why lattice-based problems matter for post-quantum cryptography.
- Interpret basic security definitions: negligible functions, indistinguishability, game-style analysis, and perfect secrecy.

Expectations from the Students

The material covered in the course is meant to give students exposure and a baseline understanding of cryptography. The topics covered here are heavily researched and deep fields: in order to cover every topic in a complete fashion would take years. However, this course is meant to serve as a robust introduction, allowing the students to experience high level topics in a simpler format than a hyper-dense mathematics course.

The level of mastery that a student can achieve over the material is directly related to the amount of time and effort they wish to put into the course. If a student shows up to the lectures and is attentive, they can expect to have a very basic understanding of the topics covered. The students will be given a large amount of supplementary material which they can choose to use to further their understanding of the topics we cover. This includes additional reading, problem sets and homework assignments.

It is imperative that before the student takes this course that they determine how much time they are willing to dedicate to their studies, as well as the level of proficiency they wish to achieve. Students who want regular feedback and direct guidance from the instructors are encouraged to consider Tier 2 or Tier 3.

Communication

- Course announcements, including the lecture platform and meeting links, will be sent **[channel: e-mail list, Signal group, website]**.
- Questions about the course, enrollment tiers, or scheduling may be sent to the instructors at the addresses above.
- Questions about the mathematics are best brought to office hours or, for Tier 3 students, to one-on-one sessions.

Important Dates

[date]	Registration opens
Mon., November 2, 2026	Week 1 begins; first lecture. Tier 2 and Tier 3 students submit their availability for office hours.
Week of November 23	Week 4; U.S. Thanksgiving falls on Thursday, November 26
Weeks of Dec. 21 and Dec. 28	Weeks 8 and 9; winter holidays
Week of Jan. 25, 2027	Week 13; final lecture

Dates are tentative. Any changes to the schedule around the holidays will be announced in advance.

Weekly Schedule

Week	Week of	Topics
1	Nov 2, 2026	The first week gives a brief overview of the course. The lecture lays out the core problems that modern cryptography seeks to solve, introduces the Diffie-Hellman Key exchange in simple terms, and dives into the mathematical notation and properties used for the rest of the course.
2	Nov 9	This week covers the property of divisibility, along with the various proof methods used in the mathematical analysis of cryptography. The lecture works through a few proofs in depth and introduces the Division Algorithm and the Euclidean Algorithm, finishing with Bezout's Lemma and the Fundamental Theorem of Arithmetic.
3	Nov 16	The course turns to modular arithmetic. The lecture formally constructs modular congruences, builds $\mathbb{Z}_n$ and its arithmetic, inspects zero divisors and units, and solves linear congruences. It closes with the Chinese Remainder Theorem and its applications in modern cryptography.
4	Nov 23	The discussion of modular arithmetic continues with how exponents behave, and uses these properties to prove the Diffie-Hellman key exchange is well formed. The lecture also examines the flaws the key exchange possesses and how it can be implemented, and finishes with the Discrete Logarithm Problem.
5	Nov 30	The course begins its inspection of Abstract Algebra. The lecture defines mathematical groups, subgroups, and cyclic groups, then defines cosets and builds up to Lagrange's Theorem.
6	Dec 7	This week defines group homomorphisms, establishes the group structure of $\mathbb{Z}_n^\times$, and proves the existence of primitive roots. The lecture then uses these ideas to analyze the Baby-Step Giant-Step algorithm as an attack, as well as Pollard's Rho Algorithm.
7	Dec 14	The lecture expands earlier definitions by introducing rings, fields, and polynomial rings. It shows how to construct certain fields, then constructs the field used in AES, and closes with a general discussion of finite fields.
8	Dec 21	This week covers Quadratic Residues and Euler's Criterion, introduces the Legendre Symbol, and presents the Tonelli-Shanks Algorithm.
9	Dec 28	The lecture studies systems of linear equations over a prime field, then introduces the Learning with Errors problem, Lattice Based problems, and the properties they possess.
10	Jan 4, 2027	This week brings in probability, introducing the Birthday Problem, the Prime Number Theorem, and Fermat's Test for Primality. The lecture then presents the Miller-Rabin test and probabilistic primes.

Week	Week of	Topics
11	Jan 11	The course defines the key notions of cryptographic analysis: negligible functions, efficient algorithms, and indistinguishability. The lecture then applies game style analysis to encryption and perfect secrecy.
12	Jan 18	The course finally takes up RSA: how it is implemented, its potential attack vectors, and how it can potentially be broken. The lecture also examines a quantum attack on RSA.
13	Jan 25	The course closes with elliptic curve cryptography: how it works, how it can be broken, and where it is implemented.

The schedule is tentative and may be adjusted as the course progresses; any change will be announced in advance.